Bar Lerer

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Lead Software Engineer at Cadence building AI for chip verification: regression learning, explainable ML, LLM agents. First-author SIAM SISC paper & ICLR 2024 workshop on deep-learning preconditioners; M.Sc. CS, GPA 95.

Experience

- **Cadence Design Systems** — Petah Tikva, Israel
  Lead Software Engineer, Verisium AI-Driven Verification Platform — *Apr 2023 - Present*
  - Built the regression-learning model for **Verisium AutoFocus**: learns from past random regressions, statistically sampling test-to-coverage-bin hit distributions to predict which tests to run
  - Implemented signal-fanout analysis for AutoFocus that traces how a design change propagates to other components, selecting efficient coverage paths through path pruning and rareness filtering
  - Developed an unreleased ML product that uncovers hidden relationships in regression data, using LightGBM with SHAP explanations to identify which factors drive outcomes
  - **Best Demo Award**, Cadence Innovation Conference Israel: built and presented an agentic code-review system, LLM agents (GitHub Copilot CLI with a custom MCP server for Perforce/Swarm) reviewing changes against domain-specific rules
  - Prototyped an agentic, RAG-based workflow that monitors git commits, classifies whether each change is user-facing, and proposes the matching documentation fix
  - Prototyped parallel regression ranking for Verisium Manager, using PCA and clustering to split the ranking computation across processes
- **The Web Information Systems Research Group (TU Delft)** — Delft, Netherlands
  Research Engineer — *Jul 2020 - Jun 2021*
  - Built Dandelion, a crowd-computing chatbot connecting TU Delft researchers with students: NLP-based personalized conversations on a RabbitMQ microservice architecture
- **KPMG** — Rotterdam, Netherlands
  Software Engineering Intern — *Apr 2020 - Jul 2020*
  - Built a privacy-focused CAPTCHA (Flask, Redis, PostgreSQL) with Docker-based CI/CD to Azure

Publications

- **Multigrid-Augmented Deep Learning Preconditioners for the Helmholtz Equation using Compact Implicit Layers**: **Lerer, B.**, Ben-Yair, I., & Treister, E. SIAM Journal on Scientific Computing 46(5), S123–S144, 2024. Also presented at the ICLR 2024 Workshop on AI for Differential Equations in Science. arXiv:2306.17486
- **An Exploratory Study on Conversational Agents Using Dynamic Conversation Styles**: Lerer, B. Bachelor's thesis, TU Delft, 2021. TU Delft Repository

Education

- **Ben-Gurion University of the Negev** — Beer Sheva, Israel
  M.Sc. in Computer Science; GPA: 95; Dean's List 2023 — *Oct. 2021 – Jul. 2023*
  - Thesis: learned preconditioner for the high-frequency Helmholtz equation, a U-Net CNN with implicit coarse-grid layers combined with multigrid; trained on grids up to 512^2 and generalized to unseen grids up to 4096^2 (16.7M unknowns)
  - Teaching Assistant: Software Engineering Foundations; Project Workshop
- **Delft University of Technology** — Delft, Netherlands
  B.Sc. in Computer Science and Engineering; Israeli GPA: 86 — *Sep. 2018 – Sep. 2021*
  - Teaching Assistant: Big Data; Software Engineering Methods; Software Project

Skills

- **ML & Data**: Python, PyTorch, scikit-learn, LightGBM, SHAP, pandas, NumPy, Cython, Numba
- **LLM Tooling**: LLM agents, RAG, MCP servers, GitHub Copilot CLI
- **Systems**: FastAPI, Docker, Linux, LSF, CI/CD, Git, Perforce